

Complete Exam-Oriented Study Notes

Introduction to Data and Statistical Foundations (Units I & II)

Source Material: Vishwakarma Institute of Technology Notes | *Target:* Comprehensive Exam Prep

UNIT I: INTRODUCTION TO DATA AND STATISTICS

1. Data Basics & Collection Principles

Data Collection Definition: The systematic method of obtaining, observing, measuring, and analyzing accurate information to support research across academic and professional domains.

Specialized Software Tools: Software platforms used for structured data collection include **KoboToolbox**, **REDCap**, **JotForm**, **Survey CTO**, and **CommCare**.

Data Matrix Layout: A standard rectangular layout where each row represents a unique **case** (or observational unit) and each column represents a **variable** (a characteristic measured for each case).

2. Classification of Data & Levels of Measurement

Data is broadly categorized into **Qualitative (Categorical)** and **Quantitative (Numerical)** data.

A. Qualitative (Categorical) Data

- **Nominal Data:** Categories with *no intrinsic order or ranking*. Changing category order does not change meaning. Allowed operations: frequencies, percentages. *Examples:* Nationality (American, Indian, German), Gender (Male, Female).
- **Ordinal Data:** Categories with a *natural ordering or ranking*, but exact mathematical distances between adjacent categories are not defined. *Examples:* Education level (Below High School, High School Diploma, Some College, Bachelor's), Dread level (Scale 1 to 5), Bedtime ranges.

B. Quantitative (Numerical) Data

- **Discrete Data:** Numerical values consisting of distinct, countable whole numbers. *Examples:* Number of migraines experienced, number of countries visited.
- **Continuous Data:** Numerical measurements along a continuous scale that can take any value within an interval. *Examples:* Hours of sleep (e.g. 5.5 hrs), water volume, height, distance.
- **Interval Data:** Measured along an equidistant scale, but *lacks a true absolute zero*. Ratios are invalid (e.g., 80°F is not 'twice as hot' as 40°F). *Example:* Temperature in Fahrenheit/Celsius.
- **Ratio Data:** Measured on an equidistant scale with a *defined absolute zero point* (zero indicates total absence). Direct ratio comparisons are valid. *Examples:* Height in feet/cm, annual income, loan amount.

3. Explanatory vs. Response Variables

- **Explanatory (Independent) Variable:** The variable suspected of affecting, predicting, or influencing another variable.
- **Response (Dependent) Variable:** The variable representing the outcome or effect evaluated.

Example 1: Evaluating if median household income impacts population change → **Income** = Explanatory, **Population Change** = Response.

Example 2: In a migraine drug trial → **Treatment group assignment** = Categorical Explanatory, **Number of migraines** = Discrete Numerical Response.

4. Observational Studies vs. Controlled Experiments

- **Observational Study:** Researchers measure or survey subjects without manipulating variables or applying treatments. *Limitation:* Can establish association/correlation, but **cannot prove causation** due to confounding variables.
- **Controlled Experiment:** Researchers actively assign subjects to treatment groups using **randomization** to isolate direct causal effects.

Stent Trial Case Study: 451 stroke patients randomly split into Treatment Group (T1: Stent + Medication) and Control Group (C1: Medication only) to measure stroke recurrence directly.

5. Sampling Techniques

- **Simple Random Sampling (SRS):** Every individual in the population has an equal probability of selection.
- **Stratified Sampling:** Population is divided into homogeneous sub-groups (**strata**) based on shared traits; a simple random sample is drawn from *every* stratum.
- **Cluster Sampling:** Population is divided into clusters; a random sample of clusters is selected, and *all individuals* in chosen clusters are sampled.
- **Multistage Sampling:** Random clusters are selected, and then a *random sample of individuals* is drawn from within each selected cluster.

Indonesian Jungle Village Case Study: To test 150 individuals for malaria across 30 remote villages, researchers select a random sample of villages (clusters) and then randomly sample individuals within those villages (Multistage Sampling).

6. Measures of Central Tendency & Dispersion

- **Central Tendency:** *Mean* (arithmetic average), *Median* (middle value, robust to outliers), *Mode* (most frequent value).
- **Dispersion:** *Range* (Max - Min), *Variance* (σ^2 or s^2), *Standard Deviation* (σ or s), *Interquartile Range* (IQR = $Q3 - Q1$).

UNIT II: PROBABILITY, RANDOM VARIABLES & DISTRIBUTIONS

1. Core Probability Concepts & Formulas

- **Random Experiment:** A process producing an uncertain outcome that cannot be predicted in advance.
- **Sample Space (S):** Set of all possible outcomes. **Event (E):** Subset of sample space.
- **Additive Rule of Probability:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. If A and B are mutually exclusive, $P(A \cap B) = 0 \rightarrow P(A \cup B) = P(A) + P(B)$.
- **Conditional Probability:** $P(A | B) = P(A \cap B) / P(B)$, provided $P(B) > 0$.
- **Bayes' Theorem:** $P(A | B) = [P(B | A) \cdot P(A)] / P(B)$.
- **Independence Test:** Events A and B are independent if $P(A | B) = P(A)$ or $P(A \cap B) = P(A) \cdot P(B)$.

2. Complete Solved Numerical Examples (Unit II)

TOPIC 1: BASIC PROBABILITY, SAMPLE SPACES & EVENTS

Example 1: Single Fair Die Roll

Problem: A single fair die is rolled. Find $P(E)$ where E = 'rolling an even number' and $P(T)$ where T = 'rolling a number greater than 2'.

Step-by-Step Solution:

- Sample Space $S = \{1, 2, 3, 4, 5, 6\}$, total outcomes = 6.
- Event $E = \{2, 4, 6\}$ (3 outcomes) $\rightarrow P(E) = 3/6 = 1/2 = 0.5$.
- Event $T = \{3, 4, 5, 6\}$ (4 outcomes) $\rightarrow P(T) = 4/6 = 2/3 \approx 0.6667$.

Example 2: Box of Marbles (With Replacement)

Problem: A box contains 10 white and 10 black marbles. Construct a sample space for drawing two marbles in succession with replacement, and find $P(\{bb\})$.

Step-by-Step Solution:

- Let w = white marble, b = black marble. Total marbles = 20.
- Sample Space $S = \{ww, wb, bw, bb\}$.
- With replacement, $P(b) = 10/20 = 1/2$.
- $P(\{bb\}) = P(b) \times P(b) = (1/2) \times (1/2) = 1/4 = 0.25$.

Example 3: Discrete Sample Space Axioms

Problem: Sample space $S = \{a, b, c, d, e\}$. Events $U = \{a, b, d\}$ and $V = \{b, c, d\}$. Given $P(a) = P(b) = 0.2$, $P(c) = P(d) = 0.1$. Find $P(e)$, $P(U)$, and $P(V)$.

Step-by-Step Solution:

- Sum of probabilities = 1 $\rightarrow P(a) + P(b) + P(c) + P(d) + P(e) = 1$.
- $0.2 + 0.2 + 0.1 + 0.1 + P(e) = 1 \rightarrow 0.6 + P(e) = 1 \rightarrow P(e) = 0.4$.
- $P(U) = P(a) + P(b) + P(d) = 0.2 + 0.2 + 0.1 = 0.5$.
- $P(V) = P(b) + P(c) + P(d) = 0.2 + 0.1 + 0.1 = 0.4$.

Example 4: Drawing a Numbered Card from a Deck

Problem: Find the probability of drawing a numbered card (cards 2 through 10) from a standard 52-card deck.

Step-by-Step Solution:

- Total cards in deck = 52.
- Numbered cards per suit (2, 3, 4, 5, 6, 7, 8, 9, 10) = 9 cards.
- Total numbered cards across 4 suits = $4 \times 9 = 36$ cards.
- $P(\text{Numbered Card}) = 36/52 = 9/13 \approx 0.6923$.

Example 5: Selecting Balls Without Replacement

Problem: A bag contains 5 green and 7 red balls. Two balls are drawn without replacement. Find $P(\text{1st Green and 2nd Red})$.

Step-by-Step Solution:

- Total initial balls = $5 + 7 = 12$.
- $P(\text{1st Green}) = 5/12$.
- Remaining balls = 11 (5 green, 6 red left after 1st green). $P(\text{2nd Red} \mid \text{1st Green}) = 7/11$.
- Joint Probability $P(G_1 \cap R_2) = (5/12) \times (7/11) = 35/132 \approx 0.2652$.

Example 6: Intersection of Events on Fair vs. Altered Die

Problem: Find $P(\text{number rolled is even AND } > 2)$ for: (1) A fair die, (2) An altered die with $P(1)=1/12$, $P(6)=3/12$, $P(2)=P(3)=P(4)=P(5)=p$.

Step-by-Step Solution:

- Target Event $E \cap T = \{4, 6\}$.
- Case 1 (Fair Die): $P(\{4, 6\}) = P(4) + P(6) = 1/6 + 1/6 = 2/6 = 1/3 \approx 0.3333$.
- Case 2 (Altered Die): $\text{Sum} = 1 \rightarrow 1/12 + 3/12 + 4p = 1 \rightarrow 4/12 + 4p = 1 \rightarrow 4p = 8/12 \rightarrow p = 2/12 = 1/6$.
- $P(\{4, 6\}) = P(4) + P(6) = 2/12 + 3/12 = 5/12 \approx 0.4167$.

Example 7: Tossing a Pair of Fair Dice (Sum = 7 or 11)

Problem: Find the probability of getting a total sum of 7 or 11 when a pair of fair dice is tossed.

Step-by-Step Solution:

- Total sample space size for 2 dice $= 6 \times 6 = 36$.
- Event A (Sum = 7): $\{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\} \rightarrow 6$ outcomes.
- Event B (Sum = 11): $\{(5,6),(6,5)\} \rightarrow 2$ outcomes.
- Mutually exclusive events $\rightarrow P(A \cup B) = 6/36 + 2/36 = 8/36 = 2/9 \approx 0.2222$.

Example 8: Contingency Table Probability

Problem: Given political party vs bond opinion joint probabilities: Party A Favors=0.12, Opposes=0.09, Undecided=0.07. Party B Favors=0.16, Opposes=0.12, Undecided=0.14.

Step-by-Step Solution:

- $P(\text{Party A}) = 0.12 + 0.09 + 0.07 = 0.28$.
- $P(\text{Favors Bond}) = 0.12 + 0.16 + 0.04 + 0.08 = 0.40$.
- $P(\text{Party A and Favors Bond}) = 0.12$.

Practice Question 1 (Added Exam Practice): Non-Disjoint Addition Rule

Problem: A single card is drawn from a 52-card deck. Find the probability of drawing an Ace or a Heart.

Step-by-Step Solution:

- $P(\text{Ace}) = 4/52$, $P(\text{Heart}) = 13/52$, $P(\text{Ace} \cap \text{Heart}) = 1/52$ (Ace of Hearts).
- $P(\text{Ace} \cup \text{Heart}) = P(\text{Ace}) + P(\text{Heart}) - P(\text{Ace} \cap \text{Heart}) = 4/52 + 13/52 - 1/52 = 16/52 = 4/13 \approx 0.3077$.

TOPIC 2: CONDITIONAL PROBABILITY & INDEPENDENCE**Example 1: Fair Die Roll (Five Given Odd)**

Problem: A fair die is rolled. Find the probability that the number rolled is 5 given that it is odd.

Step-by-Step Solution:

- Event $F = \{5\}$, Event $O = \{1, 3, 5\}$. $P(O) = 3/6 = 1/2$.
- $F \cap O = \{5\} \rightarrow P(F \cap O) = 1/6$.
- $P(F | O) = P(F \cap O) / P(O) = (1/6) / (3/6) = 1/3 \approx 0.3333$.

Example 2: Flight Departure & Arrival Operations

Problem: Given $P(\text{Arriving on time, } A) = 0.80$, $P(\text{Departing on time, } D) = 0.90$, and $P(A \cap D) = 0.75$.

Step-by-Step Solution:

- (a) $P(A | D) = P(A \cap D) / P(D) = 0.75 / 0.90 = 5/6 \approx 0.8333$.
- (b) $P(D | A) = P(A \cap D) / P(A) = 0.75 / 0.80 = 15/16 = 0.9375$.
- (c) Independence check: $P(A | D) = 0.8333 \neq P(A) = 0.80 \rightarrow$ Events are NOT independent.

Practice Question 2 (Added Exam Practice): Drawing Without Replacement

Problem: Two cards are drawn without replacement from a 52-card deck. What is $P(\text{2nd card is a King} | \text{1st card was a King})$?

Step-by-Step Solution:

- Initially 4 Kings in 52 cards.
- If 1st card is a King, 3 Kings remain out of 51 total remaining cards.
- $P(\text{2nd King} | \text{1st King}) = 3/51 = 1/17 \approx 0.0588$.

TOPIC 3: BAYES' THEOREM**Example 1: Pain Clinic Narcotic Pills & Addiction**

Problem: 10% of patients are prescribed pain pills ($P(A)=0.10$). 5% are addicts ($P(B)=0.05$). Out of patients prescribed pain pills, 8% are addicts ($P(B|A)=0.08$). If a patient is an addict, find $P(\text{prescribed pain pills, } A|B)$.

Step-by-Step Solution:

- Apply Bayes' Theorem: $P(A | B) = [P(B | A) \cdot P(A)] / P(B)$.
- $P(A | B) = (0.08 \times 0.10) / 0.05 = 0.008 / 0.05 = 0.16$ (16%).

Example 2: Clinic Liver Disease & Alcoholism

Problem: 10% of patients have liver disease ($P(A)=0.10$). 5% are alcoholics ($P(B)=0.05$). Among liver disease patients, 7% are alcoholics ($P(B|A)=0.07$). Find $P(\text{liver disease} | \text{alcoholic, } A|B)$.

Step-by-Step Solution:

- $P(A | B) = [P(B | A) \cdot P(A)] / P(B)$.
- $P(A | B) = (0.07 \times 0.10) / 0.05 = 0.007 / 0.05 = 0.14$ (14%).

Practice Question 3 (Added Exam Practice): Factory Defective Item Analysis

Problem: Machine A produces 60% of output ($P(A)=0.6$) with 2% defects ($P(D|A)=0.02$). Machine B produces 40% ($P(B)=0.4$) with 5% defects ($P(D|B)=0.05$). If an item is defective, find $P(\text{Machine A})$.

Step-by-Step Solution:

- Total Defective $P(D) = P(D|A)P(A) + P(D|B)P(B) = (0.02 \times 0.60) + (0.05 \times 0.40) = 0.012 + 0.020 = 0.032$.
- Bayes' Rule: $P(A | D) = P(D|A)P(A) / P(D) = 0.012 / 0.032 = 0.375$ (37.5%).

TOPIC 4: RANDOM VARIABLES & EXPECTED VALUES

Example 1: Expected Value of Rolling a 6-Sided Die (Discrete)

Problem: Find $E[X]$ for rolling a fair 6-sided die where X is the face value.

Step-by-Step Solution:

- $X \in \{1, 2, 3, 4, 5, 6\}$ with $P(x) = 1/6$ for each outcome.
- $E[X] = \sum x \cdot P(x) = (1+2+3+4+5+6)/6 = 21/6 = 3.5$.

Example 2: Tossing 3 Fair Coins for Number of Heads

Problem: Let X be the number of heads in 3 fair coin tosses. Find $E[X]$.

Step-by-Step Solution:

- Probabilities: $P(0)=1/8$, $P(1)=3/8$, $P(2)=3/8$, $P(3)=1/8$.
- $E[X] = 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = (0 + 3 + 6 + 3)/8 = 12/8 = 1.5$.

Example 3: Fabric Imperfections per 10 Meters

Problem: Given distribution: $x=0$ ($f=0.41$), $x=1$ ($f=0.37$), $x=2$ ($f=0.16$), $x=3$ ($f=0.05$), $x=4$ ($f=0.01$). Find $E[X]$.

Step-by-Step Solution:

- $E[X] = 0(0.41) + 1(0.37) + 2(0.16) + 3(0.05) + 4(0.01)$.
- $E[X] = 0 + 0.37 + 0.32 + 0.15 + 0.04 = 0.88$ imperfections.

Example 4: Stock Investment Expected Gain

Problem: An investor can make \$4000 profit with prob 0.3 or lose \$1000 with prob 0.7. Find expected gain.

Step-by-Step Solution:

- $E[X] = (4000)(0.3) + (-1000)(0.7) = 1200 - 700 = \500 .

Example 5: Biased Coin Tossed Twice

Problem: Coin biased such that $P(H)=3/4$, $P(T)=1/4$. Find expected number of tails X in 2 tosses.

Step-by-Step Solution:

- $P(X=0) = P(HH) = 9/16$; $P(X=1) = P(HT)+P(TH) = 6/16$; $P(X=2) = P(TT) = 1/16$.
- $E[X] = 0(9/16) + 1(6/16) + 2(1/16) = 8/16 = 0.5$.

Example 6: Mean of Continuous Uniform Distribution $U(a, b)$

Problem: Derive $E[X]$ for continuous uniform PDF $f(x) = 1 / (b - a)$ for $a \leq x \leq b$.

Step-by-Step Solution:

- $E[X] = \int [a \text{ to } b] x / (b - a) dx = [x^2 / (2(b - a))] \text{ from } a \text{ to } b$.
- $E[X] = (b^2 - a^2) / [2(b - a)] = (b - a)(b + a) / [2(b - a)] = (a + b) / 2$.

Example 7: Continuous Triangular PDF Expected Value

Problem: Find $E[X]$ for PDF $f(x) = x$ for $0 \leq x \leq 1$, and $f(x) = 2 - x$ for $1 < x \leq 2$.

Step-by-Step Solution:

- $E[X] = \int_{[0 \text{ to } 1]} x^2 dx + \int_{[1 \text{ to } 2]} x(2 - x) dx$.
- $\int_{[0 \text{ to } 1]} x^2 dx = 1/3$.
- $\int_{[1 \text{ to } 2]} (2x - x^2) dx = [x^2 - x^3/3] \text{ from } 1 \text{ to } 2 = (4 - 8/3) - (1 - 1/3) = 4/3 - 2/3 = 2/3$.
- $E[X] = 1/3 + 2/3 = 1.0$.

Example 8: Automobile Dealer Profit (Continuous)

Problem: Profit X (in \$5000 units) has PDF $f(x) = 2(1 - x)$ for $0 < x < 1$. Find average profit.

Step-by-Step Solution:

- $E[X] = \int_{[0 \text{ to } 1]} 2x(1 - x) dx = 2 \int_{[0 \text{ to } 1]} (x - x^2) dx = 2 [x^2/2 - x^3/3] \text{ from } 0 \text{ to } 1$.
- $E[X] = 2(1/2 - 1/3) = 2(1/6) = 1/3$.
- Average profit in dollars = $(1/3) \times \$5000 = \$1,666.67$.

Practice Question 4 (Added Exam Practice): Raffle Ticket Net Gain

Problem: A raffle ticket costs \$5. 1,000 tickets sold; 1 prize of \$1,000, 10 prizes of \$100. Find expected net gain.

Step-by-Step Solution:

- $P(\text{Win } \$1000, \text{ net } +\$995) = 1/1000 = 0.001$.
- $P(\text{Win } \$100, \text{ net } +\$95) = 10/1000 = 0.010$.
- $P(\text{Win } \$0, \text{ net } -\$5) = 989/1000 = 0.989$.
- $E[X] = (995)(0.001) + (95)(0.010) + (-5)(0.989) = 0.995 + 0.950 - 4.945 = -\3.00 (Loss of \$3 per ticket).

TOPIC 5: BINOMIAL DISTRIBUTION

PMF Formula: $P(X = x) = \binom{n}{x} p^x q^{n-x}$, where $q = 1 - p$. **Mean:** $\mu = np$. **Variance:** $\sigma^2 = npq$.

Example 1: Electronic Component Shock Test

Problem: Survival prob $p = 3/4$. Find probability that exactly 2 of 4 components tested survive ($n=4, x=2$).

Step-by-Step Solution:

- $n = 4, p = 3/4, q = 1/4, x = 2$.
- $P(X = 2) = \binom{4}{2} (3/4)^2 (1/4)^2 = 6 \times (9/16) \times (1/16) = 54/256 = 27/128 \approx 0.2109$.

Example 2: Rare Blood Disease Recovery

Problem: Recovery prob $p = 0.4$. 15 people contract disease ($n=15$). Find $P(X=5)$.

Step-by-Step Solution:

- $n = 15, p = 0.4, q = 0.6, x = 5$.
- $\binom{15}{5} = 3003, (0.4)^5 = 0.01024, (0.6)^{10} = 0.0060466$.
- $P(X = 5) = 3003 \times 0.01024 \times 0.0060466 \approx 0.1859$.

Example 3: Pet Insurance Buyers

Problem: 80% of buyers are women ($p=0.8$). Sample of 9 buyers ($n=9$). Find $P(\text{exactly 6 women})$.

Step-by-Step Solution:

- $n = 9$, $p = 0.8$, $q = 0.2$, $x = 6$.
- $\binom{9}{6} = 84$, $(0.8)^6 = 0.262144$, $(0.2)^3 = 0.008$.
- $P(X = 6) = 84 \times 0.262144 \times 0.008 = 0.176$ (17.6%).

Example 4: Coin Tossed 5 Times

Problem: Fair coin tossed 5 times ($n=5$, $p=0.5$). Find $P(x=2)$ and $P(x \geq 4)$.

Step-by-Step Solution:

- (a) $P(X = 2) = \binom{5}{2} (0.5)^2 (0.5)^3 = 10 / 32 = 5/16 = 0.3125$.
- (b) $P(X \geq 4) = P(X=4) + P(X=5) = \binom{5}{4} (0.5)^4 (0.5) + \binom{5}{5} (0.5)^5 = 5/32 + 1/32 = 6/32 = 3/16 = 0.1875$.

Practice Question 5 (Added Exam Practice): Industrial Quality Control

Problem: Machine has 10% defect rate ($p=0.10$). Sample of 6 items ($n=6$). Find $P(\text{at most 1 defective})$.

Step-by-Step Solution:

- $P(X \leq 1) = P(X = 0) + P(X = 1)$.
- $P(X = 0) = \binom{6}{0} (0.10)^0 (0.90)^6 = 0.5314$.
- $P(X = 1) = \binom{6}{1} (0.10)^1 (0.90)^5 = 6 \times 0.10 \times 0.59049 = 0.3543$.
- $P(X \leq 1) = 0.5314 + 0.3543 = 0.8857$ (88.57%).

TOPIC 6: NORMAL DISTRIBUTION & Z-SCORES

Standard Z-Transformation: $Z = (X - \mu) / \sigma \sim N(0, 1)$.

Example 1: Standard Normal Area Calculations

Problem: Given $Z \sim N(0, 1)$, find area: (a) To right of $z = 1.84$, (b) Between $z = -1.97$ and $z = 0.86$.

Step-by-Step Solution:

- (a) $P(Z > 1.84) = 1 - P(Z < 1.84) = 1 - 0.9671 = 0.0329$.
- (b) $P(-1.97 < Z < 0.86) = P(Z < 0.86) - P(Z < -1.97) = 0.8051 - 0.0244 = 0.7807$.

Example 2: Finding Cutoff Value k for Standard Normal

Problem: Find k such that: (a) $P(Z > k) = 0.3015$, (b) $P(k < Z < -0.18) = 0.4197$.

Step-by-Step Solution:

- (a) $P(Z < k) = 1 - 0.3015 = 0.6985 \rightarrow$ From Z-table, $k = 0.52$.
- (b) $P(Z < -0.18) = 0.4286$. Area to left of $k = 0.4286 - 0.4197 = 0.0089 \rightarrow k = -2.37$.

Example 3: Normal Probability Range

Problem: $X \sim N(\mu = 50, \sigma = 10)$. Find $P(45 < X < 62)$.

Step-by-Step Solution:

- $z_1 = (45 - 50) / 10 = -0.5$, $z_2 = (62 - 50) / 10 = 1.2$.
- $P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5) = 0.8849 - 0.3085 = 0.5764$.

Example 4: Light Bulb Lifespan

Problem: Light bulb lifespan normally distributed with $\mu = 800$ hrs and $\sigma = 40$ hrs. Find $P(778 < X < 834)$.

Step-by-Step Solution:

- $z_1 = (778 - 800) / 40 = -0.55$, $z_2 = (834 - 800) / 40 = 0.85$.
- $P(-0.55 < Z < 0.85) = P(Z < 0.85) - P(Z < -0.55) = 0.8023 - 0.2912 = 0.5111$.

Example 5: Storage Battery Lifespan

Problem: Battery life normally distributed with $\mu = 3.0$ yrs and $\sigma = 0.5$ yrs. Find $P(X < 2.3$ yrs).

Step-by-Step Solution:

- $z = (2.3 - 3.0) / 0.5 = -1.4$.
- $P(X < 2.3) = P(Z < -1.4) = 0.0808$.

Practice Question 6 (Added Exam Practice): Entrance Exam Scores

Problem: Exam scores normally distributed with $\mu = 70$ and $\sigma = 10$. Find percentage scoring above 85.

Step-by-Step Solution:

- $z = (85 - 70) / 10 = 1.5$.
- $P(Z > 1.5) = 1 - P(Z < 1.5) = 1 - 0.9332 = 0.0668$ (6.68%).

SUMMARY FORMULA SHEET

Category	Formula	Usage / Context
Probability Addition	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Union of non-disjoint events
Conditional Prob	$P(A B) = P(A \cap B) / P(B)$	Probability of A given B occurred
Bayes' Theorem	$P(A B) = [P(B A) \cdot P(A)] / P(B)$	Reversing conditional probability
Discrete Expected Value	$E[X] = \sum x \cdot P(x)$	Mean of discrete random variable
Continuous Expected Value	$E[X] = \int x \cdot f(x) dx$	Mean of continuous PDF
Binomial PMF	$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$	x successes in n independent trials
Binomial Parameters	$\mu = np$, $\sigma^2 = np(1-p)$	Mean and Variance of Binomial
Standard Z-Score	$Z = (X - \mu) / \sigma$	Transform Normal X to Standard Normal Z

IMPORTANT 5-MARK QUESTIONS & ANSWERS

Q1. Explain the four levels of measurement with suitable examples.

Problem: Explain and solve comprehensively.

Step-by-Step Solution:

1. **Nominal:** Categorical, non-ordered data (e.g. Nationality, Gender).
2. **Ordinal:** Categorical data with natural rank/order, distances undefined (e.g. Education Level, Satisfaction Rating).
3. **Interval:** Equidistant numeric scale, lacks true absolute zero (e.g. Temperature in °F/°C).
4. **Ratio:** Numeric scale with true absolute zero; ratios valid (e.g. Height, Income, Loan Amount).

Q2. Differentiate between Observational Studies and Controlled Experiments. Why can't observational studies establish causation?

Problem: Explain and solve comprehensively.

Step-by-Step Solution:

- **Observational Study:** Researchers record data without manipulating treatment conditions.
- Controlled Experiment:** Researchers randomly assign subjects to treatment vs control groups to measure direct causal impact.
- Causation Barrier:** Observational studies suffer from confounding variables and lack random treatment assignment. Hence, they show association, but **cannot establish causation**.

Q3. Compare Stratified Sampling, Cluster Sampling, and Multistage Sampling.

Problem: Explain and solve comprehensively.

Step-by-Step Solution:

- **Stratified Sampling:** Divide population into homogeneous strata; draw a simple random sample from every stratum.
- Cluster Sampling:** Divide population into clusters; randomly select a few clusters, and study *all units* in selected clusters.
- Multistage Sampling:** Select clusters randomly, and then take a *random sample of units* from within each chosen cluster.

Q4. State Bayes' Theorem and solve: In a clinic, 10% of patients get pain pills (A), 5% are addicts (B), and among pain pill patients, 8% are addicts. Find P(Pain Pills | Addict).

Problem: Explain and solve comprehensively.

Step-by-Step Solution:

- **Formula:** $P(A | B) = [P(B | A) \cdot P(A)] / P(B)$
- Given:** $P(A) = 0.10$, $P(B) = 0.05$, $P(B | A) = 0.08$.
- Calculation:** $P(A | B) = (0.08 \times 0.10) / 0.05 = 0.008 / 0.05 = 0.16$ (16%).

Q5. Define Binomial Distribution. State its conditions, parameter formulas, and calculate P(X=6) for n=9, p=0.8.

Problem: Explain and solve comprehensively.

Step-by-Step Solution:

- **Definition:** Counts successes X in n independent Bernoulli trials with constant success probability p.
- Conditions:** Fixed n, mutually exclusive outcomes (success/failure), constant p.
- Parameters:** Mean $\mu = np$, Variance $\sigma^2 = npq$.
- Calculation:** $P(X=6) = \binom{9}{6} \cdot (0.8)^6 \cdot (0.2)^3 = 84 \times 0.262144 \times 0.008 = 0.176$ (17.6%).

LAST-MINUTE REVISION SUMMARY

- **Qualitative vs Quantitative:** Qualitative = categories/names. Quantitative = numerical counts or continuous measurements.
- **Nominal vs Ordinal:** Nominal = no order. Ordinal = ranked/ordered categories.
- **Interval vs Ratio:** Interval = no absolute zero. Ratio = absolute zero present (ratios are valid).
- **Experiment vs Observational:** Experiments use *random assignment* to prove causation. Observational studies only show association.
- **Sampling:** Stratified = sample from *all* strata. Cluster = all units from *some* clusters. Multistage = sample of units from *some* clusters.
- **Conditional Probability:** $P(A | B) = P(A \cap B) / P(B)$.
- **Bayes' Theorem:** $P(A | B) = [P(B | A) \cdot P(A)] / P(B)$.
- **Binomial Distribution:** $P(X = x) = \binom{n}{x} \cdot p^x \cdot q^{n-x}$, Mean $\mu = np$, Variance $\sigma^2 = npq$.

- **Normal Standardization:** $Z = (X - \mu) / \sigma \sim N(0, 1)$.

Good luck with your exam!